

1a.

Substitution is done by turning units of plaintext into units of ciphertext, this means that knowing the exact number of 1s does not correspond to the plaintext. Knowing the number of 1s in the data does not do anything to help decrypt the cipher text. If the substitution type is a caesar shift an exhaustive search becomes possible with difficulty relative to the size of the alphabet used.

1b.

For a transposition cipher the bits are just moved around regardless of the unit size, this means the search space shrinks from 2^{64} possible plaintexts to just those that produce 5 ones. The number of potential matches is the number of permutations possible, since there are a limited number of permutations that produce exactly 5 1s. This shrinks the number of potential plaintexts to around 2^{23} which is still a lot but potentially more manageable with a high powered computer and enough time, provided the plaintext can be recognized when deciphered. The answer to part b is theoretically possible if the computer has some way to recognize the complete text and enough time to look through every possible transposition.

3.

Alexa and devices like it have a huge potential to make peoples lives easier and more productive, however there comes a significant security risk with them. Confidentiality is the first major concern as putting a microphone in the middle of a workplace can lead to exposing company secrets. These concerns will be mitigated by insisting that the echo is muted when not in use and that no recordings are saved by amazon for any purpose. The next confidentiality concern is around amazon echo being used to exfiltrate company data. This should be done by restricting any access to company info such as calendars, employee info or personal information. This information can be restricted by requiring a pin in order to access any company information. The final concern is alexa as an attack surface, to prevent this segmenting the alexa to a different network than other devices means any attacks cannot spread to other devices as well.

To ensure integrity with this integrity it is necessary to ensure that only authorized users have access to the alexa. If voice recognition is allowed to be the only form of identification it can be possible to use replay attacks or other voice attacks to impersonate authorized users, so PINs are a must. For integrity of company information it is also important to restrict access to almost all skills to those not necessary for employees.

As for Availability I would not access this technology to be a major concern as it has little ability to hamper the availability of any services that our company may offer.

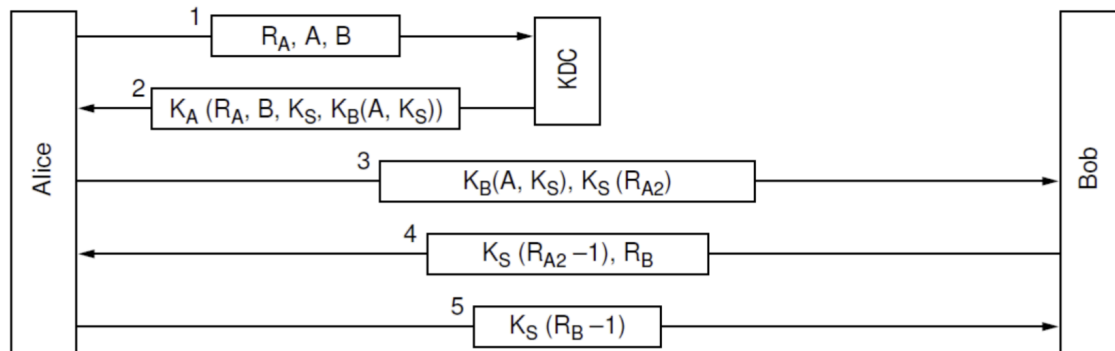
Overall these policies should keep our company safe and stop this technology from becoming a security risk. The most important things are to instate PIN usage, change privacy settings, segment the network, and mute the alexa when not in use.

4.

For usage in the home most rules stay the same except for the relaxing of a few standards. For one I think you could relax many of the rules around authentication except those around purchasing. I think using voice authentication is fine as there is a lower risk of compromise and maximizing usability is a priority. I think segmenting a home network is also optional as there are fewer sensitive devices on the network and managing segmented networks at home is difficult. I would still recommend all privacy settings such as deleting

recordings and updating privacy settings, as well as keeping the microphone muted when not in use, to maximize privacy in the home.

5.



The vulnerability in the Needham-Schroeder authentication has to do with the fact that there is no timestamp associated with Alice's first message to Bob. That means if that message is replayed later, Bob assumes that he is talking to Alice and so sends back the 4th message. From there all the attacker has to do is encrypt 1 minus the number that Bob sends back using the session key. This means that provided an attacker can acquire an old session key they can impersonate a party in the conversation. Kerberos resolves this by including timestamps and ticket lifetimes, preventing replay of old tickets. When this new layer is added Bob never responds because the replay attack registers as an incorrect request.

6. I would say that although cryptography is very important in security, I wouldn't say it's all that matters. Human networks are oftentimes just as susceptible to attack as computer networks and the exploitation of people through social engineering. I think that also many attacks come from insiders or those using credentials just as often and so securing confidentiality and integrity is often a matter of strong policy and education.